

2920/302

MANAGEMENT INFORMATION SYSTEMS

July 2020

Time: 3 hours



THE KENYA NATIONAL EXAMINATIONS COUNCIL

DIPLOMA IN INFORMATION COMMUNICATION TECHNOLOGY

MODULE III

MANAGEMENT INFORMATION SYSTEMS

3 hours

INSTRUCTIONS TO CANDIDATES

*This paper consists of **EIGHT** questions.*

*Answer any **FIVE** questions in the answer booklet provided.*

Candidates should answer the questions in English.

This paper consists of 4 printed pages.

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

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Turn over

1.
 - (a) Outline **four** challenges that an organization would face in performing frequent systems maintenance. (4 marks)
 - (b) Describe **three** components of a *management information system*. (6 marks)
 - (c) Information is a vital resource to successful management of an organization. Explain **three** roles it plays in the organization. (6 marks)
 - (d) Dan intends to protect the *intellectual property right* of an information system he developed.
 - (i) Explain the term intellectual property. (2 marks)
 - (ii) Outline **two** ways that he could use to protect this right. (2 marks)
2.
 - (a) Explain **three** reasons that would cause an information system project to fail during the implementation stage. (6 marks)
 - (b) Managers need *good information* to assist them to perform their functions in an organization. Outline **four** qualities of such information. (4 marks)
 - (c) With the aid of a diagram, describe the **three** phases of a *decision making model* in an organization. (6 marks)
 - (d) Dove Company intends to develop an information system for its operations. Outline **four** strategies that the management could put in place to ensure the success of this project. (4 marks)
3.
 - (a) Outline **four** features of *systems approach* to information systems. (4 marks)
 - (b) Describe each of the following processes as applied in management information systems:
 - (i) automation;
 - (ii) business process re-engineering;
 - (iii) total quality management. (6 marks)
 - (c) The management of Backrob Company Ltd. intends to use an information system to centralize its production and marketing operations. Explain **three** benefits the company is likely to derive from this. (6 marks)
 - (d) Organizations apply various approaches to *information systems planning*. Describe **two** such approaches. (4 marks)
4.
 - (a) Distinguish between *phishing* and *cyber stalking* as used in computer crime. (4 marks)
 - (b) A certain company intends to acquire a new human resource management system. The company has the following options to consider:
Option 1: The in-house development of the system.
Option 2: The procurement of an off the shelf software package.
 - (i) Outline **two** advantages of each of the options; (4 marks)

- (ii) Explain **one** circumstance that would necessitate the company to use each of the options. (4 marks)

- (c) Table 1 shows activities that should be undertaken in a certain project. Use it to answer the questions that follow.

Activity	Predecessor Activity	Duration (Days)
A	-	5
B	A	4
C	A	5
D	B	6
E	C	3
F	D, E	4

Table 1

- (i) Calculate each of the following for each activity:

I. total float;

II. free float.

(6 marks)

- (ii) Determine the critical path for the project.

(2 marks)

5. (a) Define each of the following terms as used in management information systems planning:

(i) policy;

(ii) vision statement;

(iii) mission statement;

(iv) objective.

(4 marks)

- (b) With the aid of a diagram, describe *SWOT analysis* technique as applied in strategic information system planning.

(6 marks)

- (c) Outline **three** ways in which *information systems* could be applied in each of the following functional areas of an organization:

(i) finance;

(ii) marketing.

(6 marks)

- (d) Distinguish between *emergency* and *routine* system maintenance techniques as applied in information systems.

(4 marks)

6. (a) Explain each of the following steps involved during the process of integrating an information systems plan to organizations' overall plan:

(i) identifying objectives and challenges;

(ii) drawing operational plan;

(iii) identifying alternatives.

(6 marks)

- (b) A certain organization has integrated an *online analytical processing* module into its management information system. Explain **three** reasons that could have prompted the management to make this move. (6 marks)
- (c) Distinguish between *dummy activity* and *event* as used in network analysis. (4 marks)
- (d) Haartz Company intends to *rationalize* their business processes and this is likely to face resistance from the employees. Outline **four** measures the management should put in place to mitigate this resistance. (4 marks)
7. (a) Define each of the following terms as used in management information system:
- (i) interaction;
- (ii) interdependence. (4 marks)
- (b) Describe **three** characteristics of an *information society*. (6 marks)
- (c) Both developed and developing countries face challenges as a result of *information society*. Explain **three** such challenges. (6 marks)
- (d) Distinguish between *throwaway* and *evolutionary* types of prototyping as applied in information systems development. (4 marks)
8. (a) (i) Describe each of the following project evaluation techniques as applied in information system project management:
- (I) internal rate of return;
- (II) return on investments. (4 marks)
- (ii) Outline **one** advantage of each of the technique in (i). (2 marks)
- (b) Explain **two** circumstances that would necessitate an organization to adopt the use of *expert systems* in its operations. (4 marks)
- (c) A certain organization has *automated* their purchasing and invoicing processes. Explain **three** benefits the organization is likely to derive from this move. (6 marks)
- (d) Galaxy Company made changes on their recently implemented management information system. Explain **two** reasons that could have led to this. (4 marks)

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